



Definite Integration & its Application

Newton-Leibnitz formula.

Let $\frac{d}{dx}(F(x)) = f(x) \forall x \in (a, b)$. Then $\int_a^b f(x) dx = \lim_{x \rightarrow b^-} F(x) - \lim_{x \rightarrow a^+} F(x)$.

- Note :** 1. If $a > b$, then $\int_a^b f(x) dx = \lim_{x \rightarrow b^+} F(x) - \lim_{x \rightarrow a^-} F(x)$.
2. If $F(x)$ is continuous at a and b , then $= F(b) - F(a)$

Example # 1 : Evaluate $\int_1^2 \frac{dx}{(x+1)(x+2)}$

Solution : $\therefore \frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$ (by partial fractions)

$$\int_1^2 \frac{dx}{(x+1)(x+2)} = [\ln(x+1) - \ln(x+2)]_1^2 = \ln 3 - \ln 4 - \ln 2 + \ln 3 = \ln \left(\frac{9}{8} \right)$$

Self practice problems :

Evaluate the following

$$(1) \int_1^2 \frac{5x^2}{x^2 + 4x + 3} dx \quad (2) \int_0^{\frac{\pi}{4}} (2\sec^2 x + x^3 + 2) dx \quad (3) \int_0^{\frac{\pi}{3}} \frac{x}{1 + \sec x} dx$$

$$\text{Ans. (1) } 5 - \frac{5}{2} \left(9\ln \frac{5}{4} - \ln \frac{3}{2} \right) \quad (2) \frac{\pi^4}{1024} + \frac{\pi}{2} + 2 \quad (3) \frac{\pi^2}{18} - \frac{\pi}{3\sqrt{3}} + 2 \ln \left(\frac{2}{\sqrt{3}} \right)$$

Property (1) $\int_a^b f(x) dx = \int_a^b f(t) dt$

i.e. definite integral is independent of variable of integration.

Property (2) $\int_a^b f(x) dx = - \int_b^a f(x) dx$

Property (3) $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, where c may lie inside or outside the interval $[a, b]$.

Example # 2 : If $f(x) = \begin{cases} x+3 & : x < 3 \\ 3x^2+1 & : x \geq 3 \end{cases}$, then find $\int_2^5 f(x) dx$.

Solution $\int_2^5 f(x) dx = \int_2^3 f(x) dx + \int_3^5 f(x) dx = \int_2^3 (x+3) dx + \int_3^5 (3x^2+1) dx = \left[\frac{x^2}{2} + 3x \right]_2^3 + \left[x^3 + x \right]_3^5$

$$= \frac{9-4}{2} + 3(3-2) + 5^3 - 3^3 + 5 - 3 = \frac{211}{2}$$





Example #3 : Evaluate $\int_2^8 |x - 5| dx$.

Solution : $\int_2^8 |x - 5| dx = \int_2^5 (-x + 5) dx + \int_5^8 (x - 5) dx = 9$

Example #4 : Show that $\int_0^2 (2x + 1) dx = \int_0^5 (2x + 1) + \int_5^2 (2x + 1)$

Solution : L.H.S. = $x^2 + x \Big|_0^2 = 4 + 2 = 6$; R.H.S. = $25 + 5 - 0 + (4 + 2) - (25 + 5) = 6$
 $\therefore$ L.H.S. = R.H.S

Self practice problems :

Evaluate the following

(4) $\int_0^4 (|x - 1| + |x - 3|) dx$

(5) $\int_{-2}^4 [x] dx$, where $[x]$ is integral part of x .

(6) $\int_0^9 [\sqrt{t}] dt$.

Ans. (4) 10 (5) 3 (6) 13

Property (4) $\int_{-a}^a f(x) dx = \int_0^a (f(x) + f(-x)) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(-x) = f(x) \text{ i.e. } f(x) \text{ is even} \\ 0, & \text{if } f(-x) = -f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$

Example #5 : Evaluate $\int_{-1}^1 \frac{3^x + 3^{-x}}{1 + 3^x} dx$

Solution : $\int_{-1}^1 \frac{3^x + 3^{-x}}{1 + 3^x} dx = \int_0^1 \left(\frac{3^x + 3^{-x}}{1 + 3^x} + \frac{3^{-x} + 3^x}{1 + 3^{-x}} \right) dx = \int_0^1 \left(\frac{3^x + 3^{-x}}{1 + 3^x} + \frac{3^x(3^{-x} + 3^x)}{1 + 3^x} \right)$
 $= \int_0^1 (3^x + 3^{-x}) dx = \left(\frac{3^x}{\ln 3} - \frac{3^{-x}}{\ln 3} \right)_0^1 = \left(\frac{3}{\ln 3} - \frac{3^{-1}}{\ln 3} \right) - \left(\frac{1}{\ln 3} - \frac{1}{\ln 3} \right) = \frac{1}{\ln 3} \left[3 - \frac{1}{3} \right] = \frac{8}{3 \ln 3}$

Example #6 : Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx$.

Solution : $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx = 2 \int_0^{\frac{\pi}{2}} \cos x dx = 2$ ($\because \cos x$ is even function)



Example # 7 : Evaluate $\int_{-1}^1 \log_e \left(\frac{2-x}{2+x} \right) dx$.

Solution : Let $f(x) = \log_e \left(\frac{2-x}{2+x} \right) \Rightarrow f(-x) = \log_e \left(\frac{2+x}{2-x} \right) = -\log_e \left(\frac{2-x}{2+x} \right) = -f(x)$
 i.e. $f(x)$ is odd function $\therefore \int_{-1}^1 \log_e \left(\frac{2-x}{2+x} \right) dx = 0$

Self practice problems :

Evaluate the following

(7) $\int_{-1}^1 |x| dx$. (8) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x dx$. (9) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec x dx}{1+2^x}$.

Ans. (7) 1 (8) 0 (9) $\ln(\sqrt{2}+1)$

Property (5) $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$. Further $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Example # 8 : Prove that $\int_0^{\frac{\pi}{2}} \frac{g(\sin x)}{g(\sin x) + g(\cos x)} dx = \int_0^{\frac{\pi}{2}} \frac{g(\cos x)}{g(\sin x) + g(\cos x)} dx = \frac{\pi}{4}$.

Solution : Let $I = \int_0^{\frac{\pi}{2}} \frac{g(\sin x)}{g(\sin x) + g(\cos x)} dx \Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{g\left(\sin\left(\frac{\pi}{2}-x\right)\right)}{g\left(\sin\left(\frac{\pi}{2}-x\right)\right) + g\left(\cos\left(\frac{\pi}{2}-x\right)\right)} dx$
 $= \int_0^{\frac{\pi}{2}} \frac{g(\cos x)}{g(\cos x) + g(\sin x)} dx$

on adding, we obtain

$$2I = \int_0^{\frac{\pi}{2}} \left(\frac{g(\sin x)}{g(\sin x) + g(\cos x)} + \frac{g(\cos x)}{g(\cos x) + g(\sin x)} \right) dx = \int_0^{\frac{\pi}{2}} 1 dx \Rightarrow I = \frac{\pi}{4}$$

Self practice problems:

Evaluate the following

(10) $\int_0^{\frac{\pi}{2}} \frac{x}{1+\sin x} dx$. (11) $\int_0^{\frac{\pi}{2}} \frac{x}{\sin x + \cos x} dx$.
 (12) $\int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$. (13) $\int_{\frac{\pi}{12}}^{\frac{5\pi}{12}} \frac{dx}{1+\sqrt{\cot x}}$

Ans. (10) π (11) $\frac{\pi}{2\sqrt{2}} \log_e(1+\sqrt{2})$ (12) $\frac{\pi^2}{16}$ (13) $\frac{\pi}{6}$





Property (6) $\int_0^{2a} f(x) dx = \int_0^a (f(x) + f(2a-x)) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$

Example #9 : Evaluate $\int_0^{\pi} \cot x \cos 2x dx$

Solution : Let $f(x) = \cot x \cos 2x$
 $\Rightarrow f(\pi-x) = \cot(\pi-x) \cos 2(\pi-x) = -\cot x \cos 2x = -f(x)$
 $\therefore \int_0^{\pi} \cot x \cos 2x dx = 0$

Example #10 : Evaluate $\int_0^{\pi} \frac{dx}{1+3\cos^2 x}$

Solution : Let $f(x) = \frac{1}{1+3\cos^2 x} \Rightarrow f(\pi-x) = f(x) \Rightarrow \int_0^{\pi} \frac{dx}{1+3\cos^2 x}$
 $= 2 \int_0^{\frac{\pi}{2}} \frac{dx}{1+3\cos^2 x} = 2 \int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{1+\tan^2 x+3} = 2 \int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{4+\tan^2 x} = \left[\tan^{-1} \left(\frac{\tan x}{2} \right) \right]_0^{\frac{\pi}{2}}$
 $\therefore \tan \frac{\pi}{2}$ is undefined, we take limit $= \lim_{x \rightarrow \pi/2-} \tan^{-1} \left(\frac{\tan x}{2} \right) - \tan^{-1} \left(\frac{\tan 0}{2} \right) = \pi/2 - 0 = \pi/2$

Example #11 : Evaluate : $\int_0^{\infty} (\cot^{-1} x)^2 dx$

Solution : Let $I = \int_0^{\infty} (\cot^{-1} x)^2 dx \Rightarrow$ Let $x = \cot \theta \Rightarrow dx = -\operatorname{cosec}^2 \theta d\theta$
 $\therefore I = \int_{\frac{\pi}{2}}^0 \theta^2 (-\operatorname{cosec}^2 \theta) d\theta \Rightarrow I = \int_0^{\frac{\pi}{2}} \theta^2 (\operatorname{cosec}^2 \theta) d\theta$
 $= (\theta^2 (-\cot \theta))_0^{\pi/2} + 2 \int_0^{\pi/2} \cot \theta d\theta \Rightarrow I = 0 + 2 \int_0^{\pi/2} \cot \theta d\theta$
 $= (2\theta \ln \sin \theta)_0^{\pi/2} - 2 \int_0^{\pi/2} \ln \sin \theta d\theta$
 $\begin{cases} \text{Standard result} \\ \int_0^{\pi/2} \ln \sin \theta d\theta = -\frac{\pi}{2} \ln 2 \end{cases} = 0 - 2 \times \left(-\frac{\pi}{2} \right) \ln 2 = \pi \ln 2.$

Self practice problems :

Evaluate the following

(14) $\int_0^{\infty} \left(\frac{\ln(1+x^2)}{1+x^2} \right) dx.$ (15) $\int_0^{\infty} \frac{\tan^{-1} x}{x(1+x^2)} dx$ (16) $\int_0^1 \ln \sin \left(\frac{\pi}{2} x \right) dx.$

Ans. (14) $\pi \ln 2$ (15) $\frac{\pi}{2} \ln 2$ (16) $-\ln 2$



Property (7) If $f(x)$ is a periodic function with period T , then

$$\begin{aligned}
 \text{(i)} \quad & \int_0^{nT} f(x) \, dx = n \int_0^T f(x) \, dx, \quad n \in \mathbb{Z} \\
 \text{(ii)} \quad & \int_a^{a+nT} f(x) \, dx = n \int_0^T f(x) \, dx, \quad n \in \mathbb{Z}, a \in \mathbb{R} \\
 \text{(iii)} \quad & \int_{mT}^{nT} f(x) \, dx = (n-m) \int_0^T f(x) \, dx, \quad m, n \in \mathbb{Z} \\
 \text{(iv)} \quad & \int_{nT}^{a+nT} f(x) \, dx = \int_0^a f(x) \, dx, \quad n \in \mathbb{Z}, a \in \mathbb{R} \\
 \text{(v)} \quad & \int_{a+nT}^{b+nT} f(x) \, dx = \int_a^b f(x) \, dx, \quad n \in \mathbb{Z}, a, b \in \mathbb{R}
 \end{aligned}$$

Example # 12 : Evaluate $\int_{-3}^5 e^{\{x\}} \, dx$, where $\{.\}$ denotes the fractional part function.

Solution :
$$\int_{-3}^5 e^{\{x\}} \, dx = (5 - (-3)) \int_0^1 e^{\{x\}} \, dx = 8 \int_0^1 e^x \, dx = 8(e^x)_0^1 = 8(e-1)$$

Example # 13 : Evaluate $\sum_{n=1}^{1000} \int_{n-1}^n |\cos 2\pi x| \, dx$

Solution :
$$\int_0^1 |\cos 2\pi x| \, dx + \int_1^2 |\cos 2\pi x| \, dx + \dots + \int_{999}^{1000} |\cos 2\pi x| \, dx = \int_0^{1000} |\cos 2\pi x| \, dx$$

Now $|\cos 2\pi x|$ is a periodic function of period $1/2$

$$I = 2000 \int_0^{1/2} |\cos 2\pi x| \, dx \Rightarrow I = 2000 \times 2 = 4000$$

Self practice problems :

Evaluate the following

(17) $\int_{-1}^{\frac{41}{2}} e^{2x - [2x]} \, dx$, where $[.]$ denotes the greatest integer function.

(18) $\int_0^{\frac{14\pi}{3}} |\sin x| \, dx$ (19) $\int_{\pi}^{\frac{3\pi}{2}} (\sin^4 x + \cos^4 x) \, dx$

Ans. (17) $\frac{43}{2}(e-1)$ (18) $\frac{19}{2}$ (19) $\frac{3\pi}{8}$

Leibnitz Theorem : If $F(x) = \int_{g(x)}^{h(x)} f(t) \, dt$, then $\frac{dF(x)}{dx} = h'(x)f(h(x)) - g'(x)f(g(x))$

Proof : Let $P(t) = \int f(t) \, dt \Rightarrow F(x) = \int_{g(x)}^{h(x)} f(t) \, dt = P(h(x)) - P(g(x))$

$$\Rightarrow \frac{dF(x)}{dx} = P'(h(x)) h'(x) - P'(g(x)) g'(x) = f(h(x)) h'(x) - f(g(x)) g'(x)$$

Example # 14 : If $F(x) = \int_x^{x^2} \sqrt{\tan t} \, dt$, then find $F'(x)$.

Solution : $F'(x) = 2x \cdot \sqrt{\tan x^2} - 1 \cdot \sqrt{\tan x}$



Example # 15 : If $F(x) = \int_{x^2}^{x^3} \frac{1}{\ln t} dt$ then find $F'(e)$

Solution : $F'(x) = \frac{3x^2}{\ln x^3} - \frac{2x}{\ln x^2} = \frac{x^2}{\ln x} - \frac{x}{\ln x} = \frac{x(x-1)}{\ln x}$ now $F'(e) = \frac{e(e-1)}{\ln e} = e(e-1)$

Example # 16 : Evaluate : $\lim_{x \rightarrow 0^+} \int_0^{x^2} \frac{\sin \sqrt{t} \tan \sqrt{t} dt}{x^4}$

Solution : Applying L' hospital rule

$$\lim_{x \rightarrow 0^+} \frac{2x \sin x \tan x}{4x^3} \Rightarrow \lim_{x \rightarrow 0} \frac{1}{2} \left(\frac{\sin x}{x} \right) \left(\frac{\tan x}{x} \right) = \frac{1}{2}$$

Example # 17 : Let $f(x) = \int_0^x (t-1)(t-2)^2 dt$, then find a point of minimum

Solution : $f(x) = \int_0^x (t-1)(t-2)^2 dt$

$$f'(x) = (x-1)(x-2)^2$$

$$\begin{array}{c} \leftarrow - \quad + \quad + \rightarrow \\ -\infty \quad 1 \quad 2 \quad \infty \end{array}$$

$\Rightarrow x = 1$ is the point of minimum

$$f(1) = \int_0^1 (t^3 - 5t^2 + 8t - 4) dt = \frac{1}{4} - \frac{5}{3} + 4 - 4 = -\frac{17}{12} \text{ . Hence } (1, -\frac{17}{12}) \text{ is a point of minimum}$$

Example # 18 : Evaluate , $\int_0^1 \frac{x^b - 1}{\ln x} dx$ 'b' being parameter.

Solution : Let $I(b) = \int_0^1 \frac{x^b - 1}{\ln x} dx \Rightarrow \frac{dI(b)}{db} = \int_0^1 \frac{x^b \ln x}{\ln x} dx + 0 - 0$ (using modified Leibnitz Theorem)

$$= \int_0^1 x^b dx = \left[\frac{x^{b+1}}{b+1} \right]_0^1 = \Rightarrow I(b) = \ln(b+1) + c$$

$$b = 0 \Rightarrow I(0) = 0 \therefore c = 0 \therefore I(b) = \ln(b+1)$$

Example # 19 : Evaluate $\int_0^1 \frac{\tan^{-1}(ax)}{x\sqrt{1-x^2}} dx$, 'a' being parameter.

Solution : Let $I(a) = \int_0^1 \frac{\tan^{-1}(ax)}{x\sqrt{1-x^2}} dx \Rightarrow \frac{dI(a)}{da} = \int_0^1 \frac{x}{(1+a^2x^2)} \cdot \frac{1}{x\sqrt{1-x^2}} dx = \int_0^1 \frac{dx}{(1+a^2x^2)\sqrt{1-x^2}}$

$$\text{Put } x = \sin t \Rightarrow dx = \cos t dt$$

$$\text{L.L. : } x = 0 \Rightarrow t = 0$$

$$\text{U.L. : } x = 1 \Rightarrow t = \frac{\pi}{2}$$

$$\frac{dI(a)}{da} = \int_0^{\frac{\pi}{2}} \frac{1}{1+a^2 \sin^2 t} \cdot \frac{1}{\cos t} \cos t dt = \int_0^{\frac{\pi}{2}} \frac{dt}{1+a^2 \sin^2 t}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2 t dt}{1+(1+a^2)\tan^2 t} = \frac{1}{\sqrt{1+a^2}} \tan^{-1} \left(\sqrt{1+a^2} \tan t \right) \Big|_0^{\frac{\pi}{2}} = \frac{1}{\sqrt{1+a^2}} \cdot \frac{\pi}{2}$$

$$\Rightarrow I(a) = \frac{\pi}{2} \ln(a + \sqrt{1+a^2}) + c \text{ But } I(0) = 0 \Rightarrow c = 0 \Rightarrow I(a) = \frac{\pi}{2} \ln(a + \sqrt{1+a^2})$$





Self Practice Problems :

(20) If $f(x) = \int_0^{x^3} \sqrt{\cos t} \, dt$, find $f'(x)$.

(21) Find the equation of tangent to the $y = F(x)$ at $x = 1$, where $F(x) = \int_x^{x^3} \frac{dt}{\sqrt{1+t^4}}$

(22) If $\int_0^x f(t) dt = x^2 - \int_x^{x^3} \frac{f(t)}{t} dt$ then find $f(1)$

(23) If $f(x) = \int_x^{x^2} x^2 \ln t \, dt$ then find $f'(e)$

(24) If $y = \int_4^{4x^2} t^4 e^{4t} dt$, Find $\frac{d^2 y}{dx^2}$

(25) If $y = \int_0^{x^2} \ln(1+t) dt$, then find $\frac{d^2 y}{dx^2}$

(26) If $\int_0^{x^2(1+x)} f(t) dt = x$ then find $f(2)$

(27) Evaluate $\int_0^\pi \ln(1+b \cos x) \, dx$, 'b' being parameter.

Ans. (20) $3x^2 \sqrt{\cos x^3}$ (21) $\sqrt{2} x - y = \sqrt{2}$ (22) $2/3$

(23) $e^2(6e-1)$ (24) $2048 e^{16x^2}$

(25) $\frac{2}{1+x^2} [2x^2 + (1+x^2) \ln(1+x^2)]$ (26) $\frac{1}{5}$ (27) $\pi \ln \left(\frac{1+\sqrt{1-b^2}}{2} \right)$

Reduction formulae in definite Integrals:

1. If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$, then show that $I_n = \left(\frac{n-1}{n} \right) I_{n-2}$

Proof : $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$

$$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x \cdot \cos^2 x \, dx = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cdot (1-\sin^2 x) \, dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \, dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x \, dx \Rightarrow I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$



Note : 1. $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$

2. $I_n = \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots I_0 \text{ or } I_1 \text{ according as } n \text{ is even or odd. } I_0 = \frac{\pi}{2}, I_1 = 1$

Hence $I_n = \begin{cases} \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots \left(\frac{1}{2}\right) \cdot \frac{\pi}{2}, & \text{if } n \text{ is even} \\ \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots \left(\frac{2}{3}\right) \cdot 1, & \text{if } n \text{ is odd} \end{cases}$

2. If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$, then show that $I_n + I_{n-2} = \frac{1}{n-1}$

Proof : $I_n = \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \cdot \tan^2 x \, dx = \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} (\sec^2 x - 1) \, dx$
 $= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \, dx = \left[\frac{(\tan x)^{n-1}}{n-1} \right]_0^{\frac{\pi}{4}} - I_{n-2}$
 $I_n = \frac{1}{n-1} - I_{n-2} \quad \therefore \quad I_n + I_{n-2} = \frac{1}{n-1}$

3. If $I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cdot \cos^n x \, dx$, then show that $I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}$

Proof : $I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^{m-1} x (\sin x \cos^n x) \, dx = \left[-\frac{\sin^{m-1} x \cdot \cos^{n+1} x}{n+1} \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \frac{\cos^{n+1} x}{n+1} (m-1) \sin^{m-2} x \cos x \, dx$
 $= \left(\frac{m-1}{n+1}\right) \int_0^{\frac{\pi}{2}} \sin^{m-2} x \cdot \cos^n x \cdot \cos^2 x \, dx = \left(\frac{m-1}{n+1}\right) \int_0^{\frac{\pi}{2}} (\sin^{m-2} x \cdot \cos^n x - \sin^m x \cdot \cos^n x) \, dx$
 $= \left(\frac{m-1}{n+1}\right) I_{m-2,n} - \left(\frac{m-1}{n+1}\right) I_{m,n} \Rightarrow \left(1 + \frac{m-1}{n+1}\right) I_{m,n} = \left(\frac{m-1}{n+1}\right) I_{m-2,n}$
 $I_{m,n} = \left(\frac{m-1}{m+n}\right) I_{m-2,n}$

Note : 1. $I_{m,n} = \left(\frac{m-1}{m+n}\right) \left(\frac{m-3}{m+n-2}\right) \left(\frac{m-5}{m+n-4}\right) \dots I_{0,n} \text{ or } I_{1,n} \text{ according as } m \text{ is even or odd.}$

$I_{0,n} = \int_0^{\frac{\pi}{2}} \cos^n x \, dx \text{ and } I_{1,n} = \int_0^{\frac{\pi}{2}} \sin x \cdot \cos^n x \, dx = \frac{1}{n+1}$

2. Walli's Formula

$$I_{m,n} = \begin{cases} \frac{(m-1)(m-3)(m-5) \dots (n-1)(n-3)(n-5) \dots}{(m+n)(m+n-2)(m+n-4) \dots} \cdot \frac{\pi}{2} & \text{when both } m, n \text{ are even} \\ \frac{(m-1)(m-3)(m-5) \dots (n-1)(n-3)(n-5) \dots}{(m+n)(m+n-2)(m+n-4) \dots} & \text{otherwise} \end{cases}$$





Example #20 : Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^2 x (\sin x + \cos x) dx$.

Solution : Given integral = $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^3 x \cos^2 x dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^3 x dx$
 $= 0 + 2 \int_0^{\frac{\pi}{2}} \sin^2 x \cos^3 x dx$ ($\because \sin^3 x \cos^2 x$ is odd and $\sin^2 x \cos^3 x$ is even)
 $= 2 \cdot \frac{1.2}{5.3.1} = \frac{4}{15}$

Example #21 : Evaluate $\int_0^{\pi} x \sin^7 x \cos^6 x dx$

Solution : Let $I = \int_0^{\pi} x \sin^7 x \cos^6 x dx$
 $I = \int_0^{\pi} (\pi - x) \sin^7(\pi - x) \cos^6(\pi - x) dx = \pi \int_0^{\pi} \sin^7 x \cos^6 x dx - \int_0^{\pi} x \sin^7 x \cos^6 x dx$
 $\Rightarrow 2I = \pi \cdot 2 \int_0^{\frac{\pi}{2}} \sin^7 x \cos^6 x dx \Rightarrow I = \pi \frac{6.4.2.5.3.1}{13.11.9.7.5.3.1} \Rightarrow I = \frac{48\pi}{9009}$

Example #22 : Evaluate : $\int_0^a x^{5/2} \sqrt{a-x} dx$

Solution : Put $x = a \sin^2 \theta \Rightarrow dx = 2a \sin \theta \cos \theta d\theta$
 Lower limit : $x = 0 \Rightarrow \theta = 0$
 Upper limit $x = a \Rightarrow \theta = \frac{\pi}{2}$.
 $\int_0^a x^{5/2} \sqrt{a-x} dx = \int_0^{\frac{\pi}{2}} 2a^4 \sin^6 \theta \cos^2 \theta d\theta = 2a^4 \times \frac{\pi}{2} \cdot \frac{(5.3.1)(1)}{8.6.4.2} = \frac{5\pi a^4}{128}$

Self Practice Problems:

Evaluate the following

(28) $\int_0^{\frac{\pi}{2}} \sin^{11} x dx$.

(29) $\int_0^{\frac{\pi}{2}} \sin^5 x \cos^4 x dx$.

(30) $\int_0^1 x^6 \sin^{-1} x dx$

(31) $\int_0^a x (a^2 - x^2)^{\frac{7}{2}} dx$.

(32) $\int_0^2 x^{3/2} \sqrt{2-x} dx$.

Ans. (28) $\frac{128}{693}$ (29) $\frac{8}{315}$ (30) $\frac{\pi}{14} - \frac{16}{245}$ (31) $\frac{a^9}{9}$ (32) $\frac{\pi}{2}$

Property (8) If $\psi(x) \leq f(x) \leq \phi(x)$ for $a \leq x \leq b$, then

$$\int_a^b \psi(x) dx \leq \int_a^b f(x) dx \leq \int_a^b \phi(x) dx$$



Property (9) If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$

Further if $f(x)$ is monotonically decreasing in (a, b) , then $f(b)(b-a) < \int_a^b f(x) dx < f(a)(b-a)$ and if $f(x)$ is

monotonically increasing in (a, b) , then $f(a)(b-a) < \int_a^b f(x) dx < f(b)(b-a)$

Property (10) $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

Property (11) If $f(x) \geq 0$ on $[a, b]$, then $\int_a^b f(x) dx \geq 0$

Example #23 : For $x \in (0, 1)$ arrange $f_1(x) = \frac{1}{9-x^2}$, $f_2(x) = \frac{1}{9-2x^2}$ and $f_3(x) = \frac{1}{9-x^2-x^3}$ in ascending order

and hence prove that $\frac{1}{6} \ln 2 < \int_0^1 \frac{1}{9-x^2-x^3} dx < \frac{1}{6\sqrt{2}} \ln 5$

Solution :

$$\begin{aligned} \because 0 < x^3 < x^2, \text{ for all } x \in (0, 1) & \Rightarrow x^2 < x^2 + x^3 < 2x^2 \\ \Rightarrow -2x^2 < -x^2 - x^3 < -x^2 & \Rightarrow 9 - 2x^2 < 9 - x^2 - x^3 < 9 - x^2 \\ \Rightarrow \frac{1}{9-x^2} < \frac{1}{9-x^2-x^3} < \frac{1}{9-2x^2} & \\ f_1(x) < f_3(x) < f_2(x) \text{ for } x \in (0, 1) & \\ \Rightarrow \int_0^1 f_1(x) dx < \int_0^1 f_3(x) dx < \int_0^1 f_2(x) dx & \\ \Rightarrow \int_0^1 \frac{dx}{9-x^2} < \int_0^1 \frac{dx}{9-x^2-x^3} < \int_0^1 \frac{dx}{9-2x^2} & \\ \Rightarrow \frac{1}{6} \left(\ln \left| \frac{3+x}{3-x} \right| \right)_0^1 < \int_0^1 \frac{dx}{9-x^2-x^3} < \frac{1}{6\sqrt{2}} \left(\ln \left| \frac{3+2x}{3-2x} \right| \right)_0^1 & \\ \Rightarrow \frac{1}{6} \ln 2 < \int_0^1 \frac{1}{9-x^2-x^3} dx < \frac{1}{6\sqrt{2}} \ln 5 & \end{aligned}$$

Example #24 : Prove that $1 < \int_0^2 \left(\frac{5-x}{9-x^2} \right) dx < \frac{6}{5}$

Solution : Let $f(x) = \frac{5-x}{9-x^2}$

$$\therefore f'(x) = -\frac{(x-9)(x-1)}{(9-x^2)^2} \Rightarrow f'(x) = 0 \text{ or not defined} \Rightarrow x = 1$$

Then $f(0) = 5/9$, $f(1) = \frac{1}{2}$, $f(2) = 3/5$ The greatest and least values of the integrand in the

interval $[0, 2]$ are respectively, equal to $f(2) = 3/5$ and $f(1) = \frac{1}{2}$

$(2-0) \frac{1}{2} < \int_0^2 \left(\frac{5-x}{9-x^2} \right) dx < (2-0) \frac{3}{5}$ Hence $1 < \int_0^2 \left(\frac{5-x}{9-x^2} \right) dx < \frac{6}{5}$





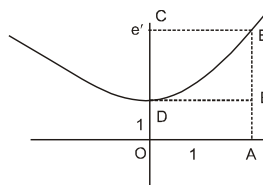
Example # 25 : Estimate the value of $\int_0^1 e^{x^2} dx$ using (i) rectangle, (ii) triangle.

Solution : (i) By using rectangle

$$\text{Area OAED} < \int_0^1 e^{x^2} dx < \text{Area OABC}$$

$$1 < \int_0^1 e^{x^2} dx < 1 \cdot e$$

$$1 < \int_0^1 e^{x^2} dx < e$$



(ii) By using triangle

$$\text{Area OAED} < \int_0^1 e^{x^2} dx < \text{Area OAED} + \text{Area of triangle DEB}$$

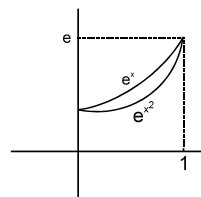
$$1 < \int_0^1 e^{x^2} dx < 1 + \frac{1}{2} \cdot 1 \cdot (e - 1) \quad 1 < \int_0^1 e^{x^2} dx < \frac{e+1}{2}$$

Example # 26 : Estimate the value of $\int_0^1 e^{x^2} dx$ by using $\int_0^1 e^x dx$.

Solution : For $x \in (0, 1)$, $e^{x^2} < e^x$

$$\Rightarrow 1 \times 1 < \int_0^1 e^{x^2} dx < \int_0^1 e^x dx$$

$$1 < \int_0^1 e^{x^2} dx < e - 1$$



Self practice problems :

(33) Prove the following : $\int_0^1 e^{-x} \cos^2 x dx < \int_0^1 e^{-x^2} \cos^2 x dx$

(34) Prove the following : $0 < \int_0^{\frac{\pi}{2}} \sin^{n+1} x dx < \int_0^{\frac{\pi}{2}} \sin^2 x dx, n > 1$

(35) Prove the following : $e^{-\frac{1}{e}} < \int_0^1 x^x dx < 1$

(36) Prove the following : $-\frac{1}{2} \leq \int_0^1 \frac{x^3 \cos x}{2+x^2} dx < \frac{1}{2}$

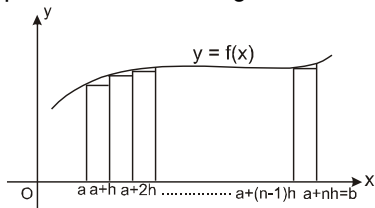
(37) Prove the following : $1 < \int_0^{\frac{\pi}{2}} \sqrt{\sin x} dx < \sqrt{\frac{\pi}{2}}$

(38) Prove the following : $\frac{4}{\pi} < \int_{\pi/4}^{\pi/3} \frac{\tan x}{x} dx < \frac{3\sqrt{3}}{\pi}$



Definite Integral as a limit of sum

Let $f(x)$ be a continuous real valued function defined on the closed interval $[a, b]$ which is divided into n parts as shown in figure.



The point of division on x-axis are $a, a + h, a + 2h, \dots, a + (n - 1)h, a + nh$, where $\frac{b-a}{n} = h$.

Let S_n denotes the area of these n rectangles.

Then, $S_n = hf(a) + hf(a + h) + hf(a + 2h) + \dots + hf(a + (n - 1)h)$

Clearly, S_n is area very close to the area of the region bounded by curve $y = f(x)$, x-axis and the ordinates $x = a, x = b$.

Hence $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} S_n$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} h f(a+rh) = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \left(\frac{b-a}{n} \right) f \left(a + \frac{(b-a)r}{n} \right)$$

Note :

1. We can also write

$$S_n = hf(a + h) + hf(a + 2h) + \dots + hf(a + nh) \text{ and } \int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{b-a}{n} \right) f \left(a + \left(\frac{b-a}{n} \right) r \right)$$

$$2. \text{ If } a = 0, b = 1, \int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{n} f \left(\frac{r}{n} \right)$$

Steps to express the limit of sum as definite integral :

Step 1. Replace $\frac{r}{n}$ by x , $\frac{1}{n}$ by dx and $\lim_{n \rightarrow \infty}$ by $\int$

Step 2. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{r}{n} \right)$ by putting least and greatest values of r as lower and upper limits respectively.

$$\text{For example } \lim_{n \rightarrow \infty} \sum_{r=1}^{pn} \frac{1}{n} f \left(\frac{r}{n} \right) = \int_0^p f(x) dx \quad \left(\because \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) \Big|_{r=1} = 0, \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) \Big|_{r=np} = p \right)$$

Example #27 : Evaluate $\lim_{n \rightarrow \infty} \left[\frac{1}{1+n} + \frac{1}{2+n} + \frac{1}{3+n} + \dots + \frac{1}{10n} \right]$

$$\begin{aligned} \text{Solution : } \lim_{n \rightarrow \infty} \left[\frac{1}{1+n} + \frac{1}{2+n} + \frac{1}{3+n} + \dots + \frac{1}{10n} \right] &= \lim_{n \rightarrow \infty} \sum_{r=1}^{9n} \frac{1}{r+n} \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^{9n} \frac{1}{n} \frac{1}{\left(\frac{r}{n} \right) + 1} = \int_0^9 \frac{dx}{x+1} = [\ln(x+1)]_0^9 = \ln 10 \end{aligned}$$





Example # 28 : Evaluate $\lim_{n \rightarrow \infty} \left[\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \frac{n+3}{n^2+3^2} + \dots + \frac{1}{n} \right]$.

Solution : $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n+r}{n^2+r^2} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \frac{1+\frac{r}{n}}{1+\left(\frac{r}{n}\right)^2} \because \lim_{n \rightarrow \infty} \left(\frac{r}{n}\right) = 0$, when $r = 1$, lower limit = 0

and $\lim_{n \rightarrow \infty} \left(\frac{r}{n}\right) = \lim_{n \rightarrow \infty} \left(\frac{n}{n}\right) = 1$, when $r = n$, upper limit = 1

$$\int_0^1 \frac{1+x}{1+x^2} dx = \int_0^1 \frac{1}{1+x^2} dx + \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx = [\tan^{-1}x]_0^1 + \left[\frac{1}{2} \log_e(1+x^2) \right]_0^1$$

$$= \frac{\pi}{4} + \frac{1}{2} \ln 2$$

Example # 29 : Evaluate : $\lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n! n^n} \right)^{\frac{1}{n}}$

Solution : Let $y = \lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n! n^n} \right)^{\frac{1}{n}} \Rightarrow \ln y = \lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\frac{(2n)!}{n! n^n} \right)$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\frac{2n(2n-1)(2n-2)\dots(n+1)}{n^n} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} [\ln(1+r/n)] = \int_0^1 \ln(1+x) dx = (x \ln(1+x))_0^1 - \int_0^1 \frac{x}{1+x} dx$$

$$= (x \ln(1+x))_0^1 - (x - \ln(1+x))_0^1 = \ln 2 - (1 - \ln 2) = \ln 4/e \Rightarrow y = 4/e$$

Self Practice Problems :

Evaluate the following limits

(39) $\lim_{n \rightarrow \infty} \left\{ \frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{1}{8n} \right\}$

(40) $\lim_{n \rightarrow \infty} \left[\frac{1}{1+n} + \frac{1}{2+n} + \frac{1}{3+n} + \dots + \frac{1}{5n} \right]$

(41) $\lim_{n \rightarrow \infty} \frac{1}{n^2} \left[\sin^3 \frac{\pi}{4n} + 2 \sin^3 \frac{2\pi}{4n} + 3 \sin^3 \frac{3\pi}{4n} + \dots + n \sin^3 \frac{n\pi}{4n} \right]$

(42) $\lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{\sqrt{n^2-r^2}}$

(43) $\lim_{n \rightarrow \infty} \left(\tan \frac{\pi}{2n} \tan \frac{2\pi}{2n} \tan \frac{3\pi}{2n} \dots \tan \frac{n\pi}{2n} \right)^{\frac{1}{n}}$

Ans. (39) $\frac{3}{8}$ (40) $\ln 5$ (41) $\frac{\sqrt{2}}{9\pi^2} (52 - 15\pi)$

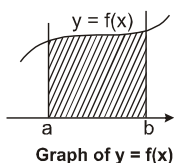
(42) $\frac{\pi}{2}$ (43) 1



Area Between The Curve :

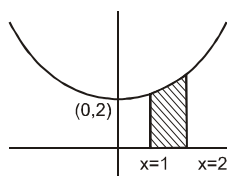
Area included between the curve $y = f(x)$, x-axis and the ordinates $x = a$, $x = b$

(a) If $f(x) \geq 0$ for $x \in [a, b]$, then area bounded by curve $y = f(x)$, x-axis, $x = a$ and $x = b$ is $\int_a^b f(x) dx$



Example #30 : Find the area enclosed between the curve $y = x^2 + 2$, x-axis, $x = 1$ and $x = 2$.

Solution :



Graph of $y = x^2 + 2$

$$\text{Area} = \int_1^2 (x^2 + 2) dx = \left[\frac{x^3}{3} + 2x \right]_1^2 = \frac{13}{3}$$

Example #31 : Find area bounded by the curve $y = \ell n x + \tan^{-1} x$ and x-axis between ordinates $x = 1$ and $x = 2$.

Solution : $y = \ell n x + \tan^{-1} x$

$$\text{Domain } x > 0, \frac{dy}{dx} = \frac{1}{x} + \frac{1}{1+x^2} > 0$$

y is increasing and $x = 1, y = \frac{\pi}{4} \Rightarrow y$ is positive in $[1, 2]$

$$\begin{aligned} \therefore \text{Required area} &= \int_1^2 (\ell n x + \tan^{-1} x) dx = \left[x \ell n x - x + x \tan^{-1} x - \frac{1}{2} \ell n(1+x^2) \right]_1^2 \\ &= 2 \ell n 2 - 2 + 2 \tan^{-1} 2 - \frac{1}{2} \ell n 5 - 0 + 1 - \tan^{-1} 1 + \frac{1}{2} \ell n 2 \\ &= \frac{5}{2} \ell n 2 - \frac{1}{2} \ell n 5 + 2 \tan^{-1} 2 - \frac{\pi}{4} - 1 \end{aligned}$$

Note : If a function is known to be positive valued then graph is not necessary.

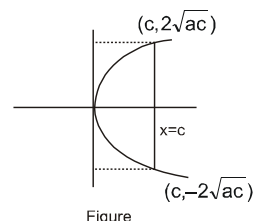
Example #32 : The area cut off from a parabola by any double ordinate is k times the corresponding rectangle contained by the double ordinate and its distance from the vertex. Find the value of k ?

Solution : Consider $y^2 = 4ax$, $a > 0$ and $x = c$

$$\text{Area by double ordinate} = 2 \int_0^c 2\sqrt{a}\sqrt{x} dx = \frac{8}{3} \sqrt{a} c^{3/2}$$

Area by double ordinate = k (Area of rectangle)

$$\frac{8}{3} \sqrt{a} c^{3/2} = k \cdot 4\sqrt{a} c^{3/2} \Rightarrow k = \frac{2}{3}$$



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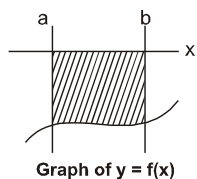
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- (b) If $f(x) < 0$ for $x \in [a, b]$, then area bounded by curve $y = f(x)$, x -axis, $x = a$ and $x = b$ is –

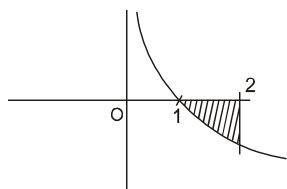
$$\int_a^b f(x) dx$$



Example # 33 : Find area bounded by $y = \log_{\frac{1}{2}} x$ and x -axis between $x = 1$ and $x = 2$

Solution : A rough graph of $y = \log_{\frac{1}{2}} x$ is as follows

$$\text{Area} = - \int_1^2 \log_{\frac{1}{2}} x dx = - \int_1^2 \log_e x \cdot \log_{\frac{1}{2}} e dx$$



$$= - \log_{\frac{1}{2}} e \cdot [x \log_e x - x]_1^2 = - \log_{\frac{1}{2}} e \cdot (2 \log_e 2 - 2 - 0 + 1) = - \log_{\frac{1}{2}} e \cdot (2 \log_e 2 - 1)$$

Note : If $y = f(x)$ does not change sign in $[a, b]$, then area bounded by $y = f(x)$, x -axis between

$$\text{ordinates } x = a, x = b \text{ is } \left| \int_a^b f(x) dx \right|$$

- (c) If $f(x) \geq 0$ for $x \in [a, c]$ and $f(x) \leq 0$ for $x \in [c, b]$ ($a < c < b$) then area bounded by curve $y = f(x)$

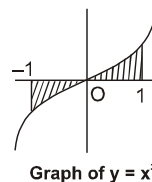
$$\text{and } x\text{-axis between } x = a \text{ and } x = b \text{ is } \int_a^c f(x) dx - \int_c^b f(x) dx$$

Example # 34 : Find the area bounded by $y = x^3$ and x -axis between ordinates $x = -1$ and $x = 1$

Solution : Required area = $\int_{-1}^0 -x^3 dx + \int_0^1 x^3 dx$

$$= \left[-\frac{x^4}{4} \right]_{-1}^0 + \left[\frac{x^4}{4} \right]_0^1$$

$$= 0 - \left(-\frac{1}{4} \right) + \frac{1}{4} - 0 = \frac{1}{2}$$



Note : Most general formula for area bounded by curve $y = f(x)$ and x -axis between ordinates $x = a$ and $x = b$

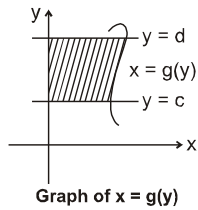
$$\text{is } \int_a^b |f(x)| dx$$




Area included between the curve $x = g(y)$, y -axis and the abscissas $y = c$, $y = d$

- (a) If $g(y) \geq 0$ for $y \in [c, d]$ then area bounded by curve $x = g(y)$ and y -axis between abscissa $y = c$ and $y = d$ is

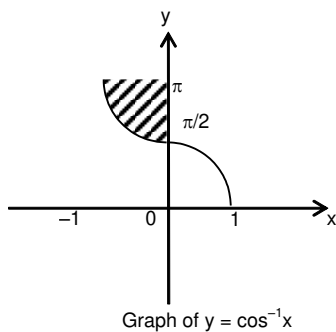
$$y = d \text{ is } \int_{y=c}^d g(y) dy$$



Example # 35 : Find area bounded between $y = \cos^{-1}x$ and y -axis between $y = \frac{\pi}{2}$ and $y = \pi$.

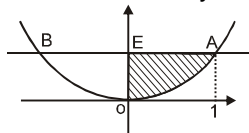
Solution : $y = \cos^{-1}x \Rightarrow x = \cos y$

$$\text{Required area} = - \int_{\pi/2}^{\pi} \cos y dy$$



$$= - \sin y \Big|_{\pi/2}^{\pi} = 1$$

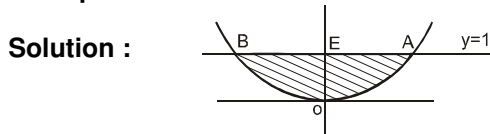
Example # 36 : Find the area bounded by the parabola $x^2 = y$, y -axis and the line $y = 1$.



Solution : Graph of $y = x^2$

$$\text{Area OEBO} = \text{Area OAEO} = \int_0^1 |x| dy = \int_0^1 \sqrt{y} dy = \frac{2}{3}$$

Example # 37 : Find the area bounded by the parabola $x^2 = y$ and line $y = 1$.



Graph of $y = x^2$

Required area is area OABO

$$= 2 \text{ area (OAEO)} = 2 \int_0^1 |x| dy = 2 \int_0^1 \sqrt{y} dy = \frac{4}{3}$$





Example # 38 : For any real t , $x = \frac{1}{2} (e^t + e^{-t})$, $y = \frac{1}{2} (e^t - e^{-t})$ is point on the hyperbola $x^2 - y^2 = 1$. Show that the area bounded by the hyperbola and the lines joining its centre to the points corresponding to t_1 and $-t_1$ is t_1 .

Solution : It is a point on hyperbola $x^2 - y^2 = 1$.

$$\begin{aligned} \text{Area (PQRP)} &= 2 \int_1^{\frac{e^{t_1} + e^{-t_1}}{2}} y dx = 2 \int_1^{\frac{e^{t_1} + e^{-t_1}}{2}} \sqrt{x^2 - 1} dx \\ &= 2 \left[\frac{x}{2} \sqrt{x^2 - 1} - \frac{1}{2} \ln(x + \sqrt{x^2 - 1}) \right]_1^{\frac{e^{t_1} + e^{-t_1}}{2}} = \frac{e^{2t_1} - e^{-2t_1}}{4} - t_1 \end{aligned}$$

$$\text{Area of } \triangle OPQ = 2 \times \frac{1}{2} \left(\frac{e^{t_1} + e^{-t_1}}{2} \right) \left(\frac{e^{t_1} - e^{-t_1}}{2} \right) = \frac{e^{2t_1} - e^{-2t_1}}{4}$$

$\therefore$ Required area = area $\triangle OPQ$ - area (PQRP) = t_1

(b) If $g(y) \leq 0$ for $y \in [c, d]$ then area bounded by curve $x = g(y)$ and y -axis between abscissa $y = c$ and $y = d$ is $-\int_{y=c}^d g(y) dy$

Note : General formula for area bounded by curve $x = g(y)$ and y -axis between abscissa $y = c$ and $y = d$ is $\int_{y=c}^d |g(y)| dy$

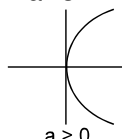
Curve-tracing :

To find approximate shape of a curve, the following phrases are suggested :

(a) **Symmetry:**

(i) **Symmetry about x-axis :**

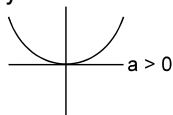
If all the powers of 'y' in the equation are even then the curve (graph) is symmetrical about the x-axis.



E.g. : $y^2 = 4ax$.

(ii) **Symmetry about y-axis :**

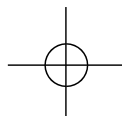
If all the powers of 'x' in the equation are even then the curve (graph) is symmetrical about the y-axis.



E.g. : $x^2 = 4ay$.

(iii) **Symmetry about both axis :**

If all the powers of 'x' and 'y' in the equation are even, then the curve (graph) is symmetrical about the axis of 'x' as well as 'y'.



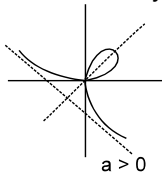
E.g. : $x^2 + y^2 = a^2$.





s (iv) Symmetry about the line $y = x$:

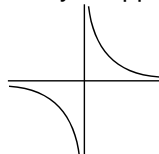
If the equation of the curve remain unchanged on interchanging 'x' and 'y', then the curve (graph) is symmetrical about the line $y = x$.



E.g. : $x^3 + y^3 = 3axy$.

(v) Symmetry in opposite quadrants :

If the equation of the curve (graph) remain unaltered when 'x' and 'y' are replaced by '-x' and '-y' respectively, then there is symmetry in opposite quadrants.



E.g. : $xy = c^2$

(b) Find the points where the curve crosses the x-axis and the y-axis.

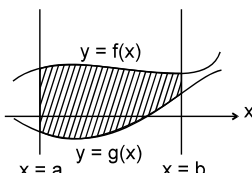
(c) Find $\frac{dy}{dx}$ and equate it to zero to find the points on the curve where you have horizontal tangents.

(d) Examine intervals when $f(x)$ is increasing or decreasing

(e) Examine what happens to 'y' when $x \rightarrow \infty$ or $x \rightarrow -\infty$

Area between two curves

If $f(x) \geq g(x)$ for $x \in [a, b]$ then area bounded by curves (graph) $y = f(x)$ and $y = g(x)$ between ordinates $x = a$ and $x = b$ is $\int_a^b (f(x) - g(x)) dx$.



Example #39 : Find the area enclosed by curve (graph) $y = x^2 + x + 1$ and its tangent at (1,3) between ordinates $x = -2$ and $x = 1$.

Solution : $\frac{dy}{dx} = 2x + 1$

$$\frac{dy}{dx} = 3 \text{ at } x = 1$$

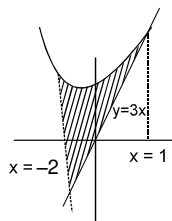
Equation of tangent is

$$y - 3 = 3(x - 1)$$

$$y = 3x$$

$$\text{Required area} = \int_{-2}^1 (x^2 + x + 1 - 3x) dx$$

$$= \int_{-2}^1 (x^2 - 2x + 1) dx = \left[\frac{x^3}{3} - x^2 + x \right]_{-2}^1 = \left(\frac{1}{3} - 1 + 1 \right) - \left[-\frac{8}{3} - 4 - 2 \right] = 9$$



Note : Area bounded by curves $y = f(x)$ and $y = g(x)$ between ordinates $x = a$ and $x = b$ is $\int_a^b |f(x) - g(x)| dx$.





Example # 40 : Find the area of the region bounded by $y = \sin x$, $y = \cos x$ and ordinates $x = 0$, $x = \pi/2$

Solution :

$$\int_0^{\pi/2} |\sin x - \cos x| dx$$

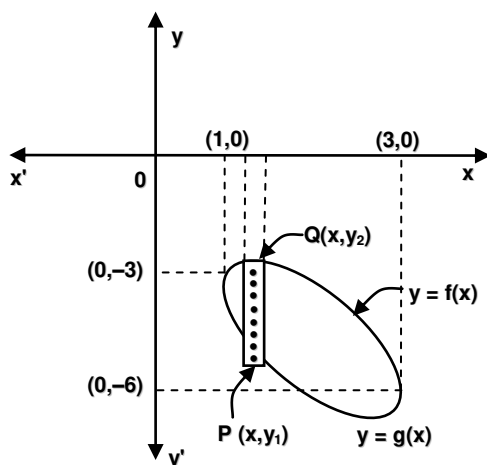
$$\int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx = 2(\sqrt{2} - 1)$$

Example # 41 : Find the area contained by ellipse $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$

Solution : $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$

$$2y^2 + 6(1+x)y + 5x^2 + 7x + 6 = 0 \Rightarrow y = \frac{-3(1+x) \pm \sqrt{(3-x)(x-1)}}{2}$$

Clearly, the values of y are real for $x \in [1, 3]$



when $x = 1$, we get $y = -3$

and, $x = 3 \Rightarrow y = -6$

Let $f(x) = \frac{-3(1+x) + \sqrt{(3-x)(x-1)}}{2}$

and, $g(x) = \frac{-3(1+x) - \sqrt{(3-x)(x-1)}}{2}$

required area = $\left| \int_1^3 \{g(x) - f(x)\} dx \right|$

$$= \left| \int_1^3 \sqrt{-x^2 + 4x - 3} dx \right| = \left| \int_1^3 \sqrt{1^2 - (x-2)^2} dx \right| = \left| \left[\frac{1}{2}(x-2) \sqrt{-x^2 + 4x - 3} + \frac{1}{2} \sin^{-1} \left(\frac{x-2}{1} \right) \right]_1^3 \right|$$

$$= \left| \left[\left\{ \frac{1}{2} \sin^{-1} 1 \right\} - \left\{ \frac{1}{2} \sin^{-1}(-1) \right\} \right] \right| = \frac{\pi}{2} \text{ sq. unit}$$





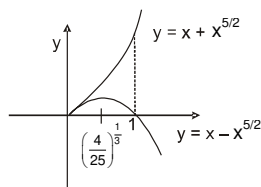
Miscellaneous examples

Example # 42 : Find the area contained between the two arms of curves $(y - x)^2 = x^5$ between $x = 0$ and $x = 1$.

Solution : $(y - x)^2 = x^5 \Rightarrow y = x \pm x^{5/2}$

For arm

$$y = x + x^{5/2} \Rightarrow \frac{dy}{dx} = 1 + \frac{5}{2} x^{3/2} > 0 \quad x \geq 0.$$



y is increasing function.

For arm

$$y = x - x^{5/2} \Rightarrow \frac{dy}{dx} = 1 - \frac{5}{2} x^{3/2}$$

$$\frac{5}{2} = 0 \Rightarrow x = \left(\frac{4}{25}\right)^{1/3}, \quad \frac{d^2y}{dx^2} = -\frac{5}{4} x^{1/2} < 0 \text{ at } x = \left(\frac{4}{25}\right)^{1/3}$$

$\therefore$ at $x = \left(\frac{4}{25}\right)^{1/3}$, $y = x - x^{5/2}$ has maxima.

$$\text{Required area} = \int_0^1 (x + x^{5/2} - x + x^{5/2}) dx = 2 \int_0^1 x^{5/2} dx = \frac{2}{7/2} x^{7/2} \Big|_0^1 = \frac{4}{7}$$

Example # 43 : Let $A(m)$ be area bounded by parabola $y = x^2 + 2x - 3$ and the line $y = mx + 1$. Find the least area $A(m)$.

Solution : Solving we obtain

$$x^2 + (2 - m)x - 4 = 0$$

Let α, β be roots $\Rightarrow \alpha + \beta = m - 2, \alpha\beta = -4$

$$A(m) = \left| \int_{\alpha}^{\beta} (mx + 1 - x^2 - 2x + 3) dx \right| = \left| \int_{\alpha}^{\beta} (-x^2 + (m-2)x + 4) dx \right|$$

$$= \left| \left(-\frac{x^3}{3} + (m-2) \frac{x^2}{2} + 4x \right) \Big|_{\alpha}^{\beta} \right| = \left| \frac{\alpha^3 - \beta^3}{3} + \frac{m-2}{2} (\beta^2 - \alpha^2) + 4(\beta - \alpha) \right|$$

$$= |\beta - \alpha| \cdot \left| -\frac{1}{3} (\beta^2 + \beta\alpha + \alpha^2) + \frac{(m-2)}{2} (\beta + \alpha) + 4 \right|$$

$$= \sqrt{(m-2)^2 + 16} \left| -\frac{1}{3} ((m-2)^2 + 4) + \frac{(m-2)}{2} (m-2) + 4 \right| = \sqrt{(m-2)^2 + 16} \left| \frac{1}{6} (m-2)^2 + \frac{8}{3} \right|$$

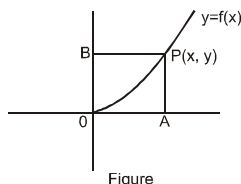
$$A(m) ((m-2)^2 + 16)^{3/2} = \frac{1}{6} \Rightarrow \text{Least } A(m) = \frac{1}{6} (16)^{3/2} = \frac{32}{3}.$$





Example # 44 : A curve $y = f(x)$ passes through the origin and lies entirely in the first quadrant. Through any point $P(x, y)$ on the curve, lines are drawn parallel to the coordinate axes. If the curve divides the area formed by these lines and coordinate axes in $m : n$, then show that $f(x) = cx^{m/n}$ or $f(x) = cx^{n/m}$ (c -being arbitrary).

Solution : Area (OAPB) = xy $\Rightarrow$ Area (OAPO) = $\int_0^x f(t) dt$



$$\text{Area (OPBO)} = xy - \int_0^x f(t) dt \Rightarrow \frac{\text{Area (OAPO)}}{\text{Area (OPBO)}} = \frac{m}{n}$$

$$n \int_0^x f(t) dt = m \left(xy - \int_0^x f(t) dt \right) \Rightarrow n \int_0^x f(t) dt = mx f(x) - m \int_0^x f(t) dt$$

Differentiating w.r.t. x

$$nf(x) = m f(x) + mx f'(x) - m f(x) \Rightarrow \frac{f'(x)}{f(x)} = \frac{n}{m} \frac{1}{x}$$

$$f(x) = cx^{n/m}$$

$$\text{similarly } f(x) = cx^{m/n}$$

Self practice problems :

- (44) Find the area bounded by the curves $y = e^x$, $y = |x - 1|$ and $x = 2$.
- (45) Compute the area of the region bounded by the parabolas $y^2 + 8x = 16$ and $y^2 - 24x = 48$.
- (46) Find the area between the x -axis and the curve $y = \sqrt{1 + \cos 4x}$, $0 \leq x \leq \pi$.
- (47) What is geometrical significance of
 - (i) $\int_0^\pi |\cos x| dx$,
 - (ii) $\int_0^{\frac{3\pi}{2}} \cos x dx$
- (48) Find the area of the region bounded by the x -axis and the curves defined by $y = \tan x$,
 (where $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$) and $y = \cot x$. (where $\frac{\pi}{6} \leq x \leq \frac{2\pi}{3}$)
- (49) Find the area bounded by the curves $x = y^2$ and $x = 3 - 2y^2$.
- (50) Find the area bounded by the curve $y = x^2 - 2x + 5$, the tangent to it at the point $(2, 5)$ and the axes of coordinates.
- (51) Find the area of the region bounded by $y = x - 1$ and $(y - 1)^2 = 4(x + 1)$
- (52) Find the area of the region lying in the first quadrant and included between the curves $x^2 + y^2 = 3a^2$, $x^2 = 2ay$ and $y^2 = 2ax$. $a > 0$



- (53) Find the area enclosed by the curves $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and the straight line $y = 3x - 15$.
- (54) Find the area bounded by the curves $4y = |4 - x^2|$, $y = 7 - |x|$
- (55) Find the area bounded by the curves $x = |y^2 - 1|$ and $y = x - 5$.
- (56) Find the area of the region formed by $x^2 + y^2 - 6x - 4y + 12 \leq 0$, $y \leq x$ and $2x \leq 5$.

- Ans.** 44. $(e^2 - 2)$ sq. units 45. $32\sqrt{\frac{2}{3}}$ sq. units
46. $2\sqrt{2}$ sq. units
47. (i) Area bounded by $y = \cos x$, x-axis between $x = 0$, $x = \pi$.
 (ii) Difference of area bounded by $y = \cos x$, x-axis between $x = 0$, $x = \frac{\pi}{2}$ and area bounded by $y = \cos x$, x-axis between $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$.
48. $\ln \frac{3}{2}$ 49. 4 sq. units
50. $8/3$ sq. units
51. $64/3$ sq. units 52. $a^2 \left[\frac{\sqrt{2}}{3} + \frac{3}{2} \sin^{-1} \frac{1}{3} \right]$ sq. units 53. $\frac{73}{6}$ sq. units
54. 32 sq. units 55. $\frac{109}{6}$ sq. units 56. $\left(\frac{\pi}{6} + \frac{1}{8} - \frac{\sqrt{3}}{8} \right)$ sq. units